

The \$20 Loading Block

There is a real gap between the \$9 plastic tray and the \$55 billet aluminum one. Whether a business fits inside it depends almost entirely on one decision he hasn't made yet — and it isn't price.

Prepared for Spencer · product: precision reloading tray · stated plan: 3D printed, \$20-40, direct-to-consumer

THE SHORT VERSION

The product may well be superior. The *plan* — 3D printed, \$20-40, DTC — is the weakest of the four available paths, and the reasons are structural rather than something better marketing fixes.

Two of the three variables fight each other. He picked **material and build quality** as the edge, and **3D printing** as the process. In this specific market, printed plastic is the commodity tier — Etsy is saturated with it and free STL files for loading blocks are on their fourth community revision. So the claim is being made in the one material the buyer already distrusts, at a price above the material he trusts less but costs less, in a process a meaningful share of his customers own outright.

Fix that contradiction and there is a business here. Leave it and there's a hobby that loses money slowly.

01 How big this actually is

Every syndicated report you'll find sizes *reloading equipment* — presses, dies, tumblers, progressive machines. None of them break out trays, and the headline numbers are the least useful thing in this analysis. They're here so nobody quotes them at him later without the caveat.

FIGURE	SOURCE POSITION	CONFIDENCE
\$2.8B → \$4.7B by 2034	Global reloading equipment, 5.9% CAGR	ESTIMATE
\$2.91B by 2032	Same category, different house, 6.87% CAGR	ESTIMATE
~44% / ~\$1.24B	North America share, US-dominated	ESTIMATE
1%–10% of gun owners	Share who handload. No registry exists.	UNKNOWN

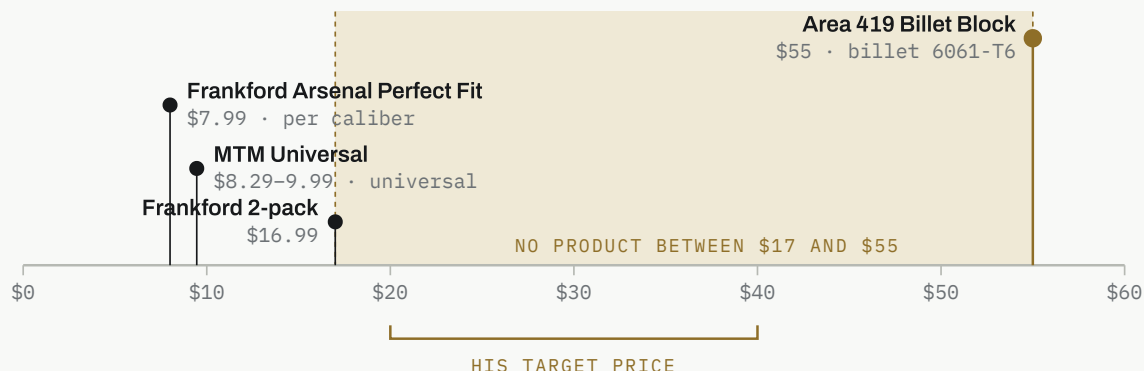
Two reports on the same category disagree by a full point of CAGR and eight hundred million dollars. Treat them as directional only: **the category is real, US-centric, and growing single digits.** That's all they support.

The number that actually matters

A loading block is bought roughly once or twice in a reloading lifetime, per case family. There is no replacement cycle, no consumable, no subscription. That single fact caps the whole opportunity, and no market report will tell you it. Here's the arithmetic, shown so it can be argued with:

- Assume 2 million US handloaders. **ESTIMATE**
- Assume 10% buy a block in a given year → **200,000 units.**
- At a \$15 blended realized price → a ~\$3M/year US retail category. At \$25, ~\$5M.
- A strong new entrant takes 2–5% inside three years → **\$60k–\$250k of annual revenue.**

That is a genuinely good side business and a bad venture. Set the expectation there before anyone spends money on tooling, and the rest of the decisions get easier.



Prices pulled from manufacturer and retailer listings on 1 September 2026. The Area 419 review count and rating are from its own product page and are the best available proxy for real volume at the premium end. Sinclair's polyethylene blocks sit in the commodity tier at Brownells; Lyman and RCBS universal trays sit alongside MTM around \$10–15.

02 Why the middle is empty

The instinct on seeing that chart is that nobody has thought of the middle. They have. The middle is empty because it's hard to defend from both sides at once:

- **From below** — at \$25 the casual reloader asks *why not the \$9 one?* He is buying a thing to stop brass falling over. Nine dollars already does that.
- **From above** — at \$25 the reloader who *cares* asks *why not the \$55 aluminum one?* He has already decided this is a tool he'll own for thirty years, and \$30 more is nothing against that.

So the \$20–40 tier needs a reason to exist that is neither "cheaper than Area 419" nor "nicer than MTM." **Quality is not that reason.** Quality is what the \$55 block already sells, and it sells it in a material printed plastic can't argue with.

A different *function* is that reason. If the tray does something neither the \$9 nor the \$55 block does, the middle stops being a compromise and becomes its own category. That's the whole ballgame, and section 04 is where the evidence for what that function could be lives.

03 The 3D printing problem

This is the hardest part of the analysis and the part most worth him hearing before he spends anything. Three issues compound.

ISSUE A — PRINTING IS THE COMMODITY HERE, NOT THE PREMIUM

Etsy has auto-generated category pages for "3d printed reloading tray," "3d printed reloading block," and "3d printed reloading accessories." Those pages only exist when listing density is high. Sellers ship PLA by default with PETG or ABS on request, in multiple colors, often with shipping included, in roughly the \$10–25 band.

His \$20–40 target therefore doesn't sit *above* the printed tier. It sits **inside the top of it**, competing on price with hobbyists who have no overhead and no ambition to build a brand.

ISSUE B — THE FREE FILES ALREADY EXIST, AND THEY'RE MATURE

Printables hosts a *Customizable Stacking Loading Block V4* — a fourth-revision remix of an earlier Thingiverse design — and a *Universal Loading Tray* the author has tested against .308, .223, 9mm, 7.65 Para and .45 Auto in 20, 30, 40 and 50-round capacities. Cults3D lists 37 free reloading-tray models. MakerWorld and Thingiverse carry more.

The overlap between people who handload and people who own a 3D printer is unusually high — both are tinkering hobbies about making your own version of a manufactured thing. **A meaningful share of his addressable market can print a competent competitor for about a dollar fifty in filament.**

ISSUE C — THE CLAIM AND THE PROCESS CONTRADICT EACH OTHER

He named material and build quality as the edge. Against \$55 billet 6061-T6, an FDM print doesn't compete. Against a \$9 injection-molded tray it isn't obviously better either: PLA's glass transition is around 140°F, which a closed garage or a truck bed clears in a Washington August, and PLA is hygroscopic — it takes on moisture and goes brittle.

Worse, the forums are explicit that the standing complaint about plastic blocks is exactly this. *"Plastic reloading blocks wear out, get loose, and build static."* He'd be making a material argument inside the one material category the market has already convicted.

None of this says the tray is bad. It says **printing is the wrong way to sell it** — and, crucially, it's fine as the way to *prove* it. That distinction drives the recommendation.

04 What reloaders actually complain about

This is the useful half. The complaints across Sniper's Hide, AccurateShooter, The Firearms Forum and the manufacturer literature are consistent, specific, and — notably — **not about material**. They are about geometry.

THE FAILURE	WHAT IT COSTS THE RELOADER
Cases sit too deep in universal trays	The powder funnel won't seat on the case mouth. He has to lift each case out, funnel it, and set it back — which defeats the entire point of a tray. Repeatedly cited.
Holes too shallow	Cases tip when the block is flipped or carried across the bench. A tipped charged case is spilled powder at best; a light charge you don't notice at worst.
Cartridges wobble	Universal trays fit one case family well and the rest loosely. Long or heavy brass is worst.
Holes too tight	Users report drilling out MTM trays because cases stick and lift the whole tray off the bench when withdrawn.
Static	Plastic blocks hold static; loose powder granules cling where they shouldn't.
Inconsistent pocket depth in cheap aluminum	Budget machined blocks vary hole to hole, producing wobble the buyer paid a premium specifically to avoid.

Decode what \$55 is actually buying

Area 419's block is worth studying closely because 235 reviewers at 4.93 stars tell you what the premium buyer is paying for, and it isn't the metal. The metal is *how*, not *why*. What the listing and reviews emphasize:

- Holes bored to 0.500" deep with a tight fit — *"no chance of tipping"*
- Seven bored diameters offered, from 0.390" (.223) to 0.655" (CheyTac)
- A small divot in each hole to **catch spilled powder before it reaches a primer**
- Rubber pads underneath — *"does not move"*
- An integrated bullet tray so seated bullets have somewhere to live

That's four or five separate problems solved in one object. The price follows from the problem count, not the material.

The question to put to your brother

Not "why is it better." Which of the failures in that table does his tray eliminate, and can he demonstrate it on video in twenty seconds, side by side against an MTM?

If the answer is *it's just nicer*, there's no business at any price. If the answer is *it's the only tray where the powder funnel seats correctly on every case from .223 through belted magnum without swapping trays or adapters — watch*, then there is one, and the middle of that price ladder is exactly where it belongs.

One caution on the word "precision"

A loading block does not affect group size. It holds brass upright. Claiming a precision or concentricity benefit is the kind of overclaim this specific audience punishes hardest — handloaders are close to the most claim-skeptical consumers in America, and one forum regular calling it marketing nonsense travels further than any ad he could buy.

The honest and genuinely valuable framing is **error prevention**: no tipped case, no spilled charge, no missed double-charge, funnel seats square every time. That's defensible, it's what the \$55 block sells, and it's true.

05 The channel walls

"Build a DTC brand" is the part of the plan with the most specific, least negotiable obstacles. Anything firearms-adjacent runs into platform policy immediately.

CHANNEL	STATUS	DETAIL
Meta ads	CLOSED	Meta prohibits ads promoting the sale of weapons, ammunition or explosives, and treats reloading components as ammunition. Accessory ads are restricted. Enforcement is leaky — investigations found 230+ violating ads approved — but a leaky ban isn't a growth channel. When it catches up it takes the account, the pixel and the ad history with it.
Google & YouTube ads	CLOSED	Firearms and accessory promotion is restricted across Google properties.
Payments	FRICTION	Shopify Payments doesn't permit firearm/ammo-adjacent sales; Stripe, PayPal and Square routinely shut tactical merchants down. The workaround is real — Authorize.Net paired with an individually underwritten high-risk merchant account — but it costs more per transaction and adds setup. On a \$25 item, that spread matters.

CHANNEL	STATUS	DETAIL
Amazon	OPEN	Amazon bans ammunition and components — primers, powder, projectiles — but explicitly permits reloading kits and equipment . A tray is a tool, not a component. That makes Amazon Search a working paid channel in a category where Meta and Google are not. Unusual, and the single most valuable channel fact in this analysis.
Dealer / distribution	OPEN, THIN	Brownells, MidwayUSA, Midsouth, Graf's, Natchez and Powder Valley all stock loading blocks. Wholesale runs roughly 50% off retail — at \$25 retail that leaves about \$12.50 to cover a printed part, packaging and shipping. Brutal on print economics, workable on molded.
Earned / community	THE REAL ONE	Forum threads, reloading YouTube channels, and presence at precision matches. This is how Area 419 was built and it's the only channel not blocked by policy. It's free, it's slow, and it only works if the product is genuinely remarkable — which is a feature, not a bug, because it filters out the mediocre version of this plan early.

06 Defensibility

A design patent covers appearance only; a utility patent covers function. Both cost money and months. Neither is enforceable against the failure mode that actually matters here — someone measures the tray, models it, and posts a free STL. The IP literature is blunt that chasing individual printers isn't economic.

Practical moats, best to worst:

- **Brand and review depth on Amazon.** Real, and precisely why Frankford and MTM own the low end. Reviews compound; a printed file doesn't.
- **A process a home printer can't replicate.** Injection-molded geometry with undercuts, an overmolded rubber base, a machined pocket. This is the strongest structural moat available.
- **A system, not a part.** Tray plus funnel plus lid plus transport case that only work properly together. Harder to clone piecemeal.
- **Patents.** Last, and only worth filing if the mechanism is genuinely novel.

Which puts the decisive point plainly: **3D printing is the one production method with no moat at all in a market full of 3D printer owners.**

1

Prove it printed, then tool it RECOMMENDED

Treat the printed units as what they are — a working prototype and a market test, not the product. Sell 100–200 at \$25–30 through Etsy and forum classifieds, concentrating on the calibers where existing options are worst. Send free units to ten reloading YouTubers and ten forum regulars.

The objective isn't profit. It's three things: proof people pay, the exact words they use for *why*, and a review base. If 200 move in 90 days and the reviews keep naming the same specific failure it solves, get injection molding quoted. Tooling for a part this size typically runs \$8k–25k depending on undercuts ESTIMATE, and unit cost drops to roughly \$1.50–4. That's the version with real margin at \$25 and a moat a printer can't cross.

If it doesn't move 200, he's out a few hundred dollars of filament instead of twenty thousand in steel.

2

Go up, not down STRONG ALTERNATIVE

If the edge really is build and case control, the honest expression of it is machined aluminum at \$50–70, competing directly with Area 419 — where the buyer already accepts the price, isn't price-shopping, and where printed plastic has zero credibility. Higher unit cost, far better margin, much smaller volume, no Amazon race to the bottom, no tooling risk.

This is the better business *if* his tray genuinely beats the Area 419 block on something specific. Worth answering honestly before defaulting to path 1.

3

License it to a major

Realistically low odds. Hornady, RCBS, Lyman and Frankford rarely license a simple molded accessory they could design around in an afternoon. The pitch only lands with an issued utility patent on a genuinely novel mechanism, and even then the royalty on a \$10 accessory is small. Worth a letter once there are sales and reviews to point at. Not worth planning around.

4

DTC brand on printed parts at \$20–40

THE STATED PLAN

This puts a no-moat product at a price point squeezed from both sides, inside the exact bracket where free STLs and zero-overhead Etsy sellers already live, with paid social closed and payments obstructed. Every individual piece of it is survivable. All of them together is not a business — it's a job that pays under minimum wage once print time is counted.

08 What to find out before spending anything

01 What does the tray do that a \$9 MTM and a \$55 Area 419 both fail at?

One sentence, demonstrable on camera. Everything else in this document is downstream of that answer.

02 Print cost and print time per unit. At three hours a unit, one printer caps out around 200–250 a month before it stops being a business and becomes a shift.

03 What material, specifically? If it's PLA, the heat and brittleness objections are real and will surface in reviews within one summer. PETG or ABS at minimum; that changes cost and print time.

04 Get five strangers to use one — r/reloading, Sniper's Hide, AccurateShooter — and ask what they'd pay *before* telling them the price. Unprompted pricing from a stranger is worth more than any of the market figures in section 01.

05 Pull real Amazon review counts on the top five trays with Helium 10 or Jungle Scout. Review count over time is the cleanest available proxy for category volume and will settle the sizing question better than any syndicated report.

06 Price an injection mold now, not later. A quote costs nothing and it sets the real bar the printed test has to clear.

Confidence: competitor prices, platform policies and the free-STL landscape were verified directly on 1 September 2026 **KNOWN**. Category size is syndicated and the two available reports disagree materially **ESTIMATE**. The US handloader population has no authoritative source **UNKNOWN**. The unit math in section 01 and the tooling range in path 1 are my own arithmetic, shown in full so they can be argued with.

[Area 419 — Billet Loading Block, specs, price and 235 reviews](#)

[Frankford Arsenal — Perfect Fit Reloading Trays](#) · [MTM Universal Loading Tray](#) · [Sinclair poly blocks at Brownells](#)

[K&M Precision — reloading block systems](#) · [21st Century Innovation — machined loading blocks](#)

[Printables — Customizable Stacking Loading Block V4](#) · [Printables — Universal Loading Tray](#) · [Cults3D — 37 free reloading tray models](#)

[Etsy — 3D printed reloading tray category](#) · [Etsy — 3D printed reloading accessories](#)

[RCBS — Reloading Blocks Explained](#) · [Sniper's Hide — loading block discussion](#) · [AccurateShooter — best reloading block](#)

[Tech Transparency Project — Meta firearms ad policy and enforcement](#) · [Amazon restricted products — reloading equipment permitted](#) · [Shopify payment gateways for firearms-adjacent merchants](#)

[Verified Market Reports — reloading equipment market size](#) · [GlobeNewswire — competing category forecast](#)

[Design patents for 3D printed products](#) · [Haug Partners — IP and 3D printing enforcement limits](#)